

4. A researcher studied the effects of three experimental diets with varying fat contents on the total lipid (fat) level in plasma. Total lipid level is a widely used predictor of coronary heart disease. Fifteen male subjects who were within 20 percent of their ideal body weight were grouped into five blocks according to age. Within each block, the three experimental diets were randomly assigned to the three subjects. Data on reduction in lipid level (in grams per liter) after the subjects were on the diet for a fixed period of time follow.

		Fat Content of Diet		
Block <i>i</i>		<i>j</i> = 1	<i>j</i> = 2	<i>j</i> = 3
		Extremely Low	Extremely Low	Moderately Low
1	Ages 15-24	.73	.67	.15
2	Ages 25-34	.86	.75	.21
3	Ages 35-44	.94	.81	.26
4	Ages 45-54	1.40	1.32	.75
5	Ages 55-64	1.62	1.41	.78

- Why do you think that age of subject was used as a blocking variable?
- Obtain the analysis of variance table.
- Prepare a normal probability plot of the estimated treatment means. Does it appear that the treatment means differ substantially here?
- Estimate $D_1 = \mu_{.1} - \mu_{.2}$ and $D_2 = \mu_{.2} - \mu_{.3}$ using the Bonferroni procedure with a 95 percent family confidence coefficient. State your findings.